

ArogyaBandhu (ಆರೋಗ್ಯ ಬಂಧು)

Complete Technical & Operational Architecture Plan | 100% Offline-First AI Tele-Triage Platform

✦ PROBLEM STATEMENT IN ONE LINE

Government of Maharashtra (PS 133): Rural communities lack accessible, quality public healthcare due to travel barriers, offline connectivity blackouts, language gaps, fragmented records, and delayed referrals.

💡 OUR SOLUTION IN ONE LINE

ArogyaBandhu is an **offline-first, bilingual (Kannada + English), AI-triage platform** that empowers frontline ASHA workers to detect high-risk maternal, child, and chronic cases at the doorstep, generate instant QR referral slips, and seamlessly connect with PHC medical officers without requiring internet connectivity.

🎯 The 14 Official PS Features — Numbered Matrix

Direct mapping of all features stipulated in the official SIH 2026 Problem Statement 133, indexed with feature tags [1] to [14] used across the operational flow.

#	OFFICIAL FEATURE REQUIREMENT	AROGYABANDHU IMPLEMENTATION
[1]	Assisted teleconsultation	PHC Doctor tele-consult portal with video/audio triage support linking ASHA & patient.
[2]	Appointment & queue management	Smart PHC doctor referral queue sorted dynamically by clinical severity (● Red cases top).
[3]	Digital triage	Evidence-based algorithmic triage on vitals → instant ● High, ● Medium, ● Low risk scoring.
[4]	Longitudinal patient records	Persistent patient medical history stored locally in IndexedDB and synchronized with cloud.
[5]	Referral tracking	Instant QR referral slip with live state machine tracking (Referred → In Transit → Acted).
[6]	Diagnostic coordination	Automated high-risk alert routing to PHC Doctor with complete symptom/vitals breakdown.
[7]	Medicine availability	Real-time drug inventory tracker for essential items (IFA, ORS, Paracetamol) across PHCs.
[8]	High-risk patient follow-up	Post-care follow-up scheduler sending reminder notifications to ASHA field workers.
[9]	Facility dashboards	Taluk Health Officer (THO) district analytics heatmap, caseload trends, and resource alerts.
[10]	Frontline worker support	Ergonomic ASHA Field App tailored for high-speed offline data entry on entry-level phones.
[11]	Low-connectivity environments	100% offline-first architecture with automatic background synchronization on network reconnect.
[12]	Multilingual interaction	Instant 1-click bilingual toggle across all screens (ಕನ್ನಡ ↔ English).
[13]	Emergency escalation	One-touch "108 Arogya Kavacha" ambulance dispatch simulation for critical cases.
[14]	Interoperable health records	Native 14-digit ABHA ID integration conforming to ABDM (Ayushman Bharat Digital Mission).

Solution Architecture — Three Portals, One Integrated System

PORTAL 1 • MOBILE VIEW

ASHA Field App

Primary Users: ASHA & ANM frontline workers at village households.

Key Capabilities:

- 100% Offline Triage [3]
- Local Patient Records [4]
- QR Slip Generator [5]
- Maternal / Child Modules [10]
- Kannada Audio/Text [12]

PORTAL 2 • DESKTOP VIEW

PHC Doctor Portal

Primary Users: Medical Officers & staff at Primary Health Centres.

Key Capabilities:

- Teleconsultation Desk [1]
- Triage Prioritized Queue [2]
- Referral Status Update [5]
- 108 SOS Dispatch [13]
- Drug Stock Monitor [7]

PORTAL 3 • ANALYTICS VIEW

THO District

Dashboard

Primary Users: Taluk & District Health Officers (Admin).

Key Capabilities:

- Taluk Heatmap & Trends [9]
- Medicine Stock Alerts [7]
- Referral Quality Audit [8]
- High-Risk Clusters [9]
- ABHA ID Tracking [14]

End-to-End Operational Workflow (Phases 1 to 4)

PHASE 1: Frontline Village Visit (Zero Connectivity)

Scenario: ASHA worker Shantha Devi visits Goravanahalli village, Koratagere Taluk, Tumakuru. Her phone has zero network signal.

Step 1: Open Offline Application

[11] [12]

- App opens instantly from local cache (PWA architecture) without requesting internet.
- Prominent banner indicates: "🔴 **Offline Mode — Data will auto-save on your device**".
- Interface loads in ಕನ್ನಡ (Kannada) with 1-tap English toggle.

Step 2: Patient Registration & Longitudinal Record Search

[4] [14]

- Enters Name, Age, Village, Phone, and 14-digit ABHA ID.
- If patient exists in local IndexedDB, previous visits auto-populate (e.g., "Last visit: 14 Sep — Hb: 9.2 g/dL, Moderate Anaemia").

Step 3: Dedicated Clinical Triage Module Selection

[3] [10]

Module A: Maternal ANC

Gestational week, BP, Hb levels, pedal edema, severe headache, vaginal bleeding, fetal kick count.

Module B: Child Health (0–5y)

Age (months), weight, MUAC arm tape (🔴/🟡/🟢), chest indrawing, lethargy, fever days, vaccination.

Module C: NCDs / Infections

Random blood sugar, BP, persistent cough (>2 weeks), chest pain, fever, weight loss, night sweats.

Step 4: On-Device AI Triage Evaluation

[3]

● **HIGH RISK – URGENT REFERRAL RECOMMENDED** Patient: Lakshmi Bai (ಲಕ್ಷ್ಮಿ ಬಾಯಿ) | 34 Weeks Pregnant ⚠ Red Flags Detected: • Blood Pressure: 162/104 mmHg (Severe Pre-eclampsia Alert) • Hemoglobin: 7.2 g/dL (Severe Maternal Anaemia) • Symptom: Persistent facial edema & blurred vision → Action: Refer to Koratagere PHC immediately. Risk of maternal mortality.

Step 5 & 6: QR Slip Generation & Offline Encrypted Storage

[4]

[5]

[11]

- Generates a dynamic **QR Referral Slip** containing clinical summary and unique Referral ID.
- Saves complete encounter to IndexedDB queue: *"1 record pending synchronization"*.
- Generates SMS / WhatsApp text slip for the patient's family to present at the PHC gate.

PHASE 2: Network Reconnect & Automated Cloud Synchronization

Step 7: Seamless Auto-Sync & Real-Time Alert Dispatch

[6]

[11]

- ASHA moves near PHC or catches 2G/Wi-Fi signal.
- App automatically executes background batch push: "✅ **Synced 3 offline records with Tumakuru Health Cloud**".
- Real-time websocket/push triggers red alert on Koratagere PHC Medical Officer's console.

PHASE 3: PHC Doctor Consultation & Emergency Action

Step 8 & 9: Triage Prioritized Queue & Clinical Intervention

[1]

[2]

[6]

[13]

INCOMING REFERRAL QUEUE – KORATAGERE PHC -----
----- ● **1. Lakshmi Bai (Goravanahalli) – 34w ANC • BP 162/104 • Hb 7.2 [25m ago]** ASHA: Shantha Devi (#104) | [\[View Record\]](#) [\[Start Tele-Consult\]](#) [\[Dispatch 108 Ambulance\]](#) ● **2. Saraswathi M. (Nittur) – 26w ANC • BP 138/88 • Moderate Anaemia [2h ago]** ASHA: Geetha Kumari (#015) | [\[View Record\]](#) [\[Schedule Appointment\]](#) [\[Issue e-Prescription\]](#)

- **Red Case Action:** Doctor initiates Assisted Teleconsultation [\[1\]](#), triggers **108 Arogya Kavacha SOS Dispatch** [\[13\]](#), and updates status to *"Ambulance En Route"* [\[5\]](#).
- **Yellow Case Action:** Doctor schedules appointment for 2 days later [\[2\]](#) and e-prescribes IFA tablets [\[6\]](#).

Step 10 & 11: Drug Inventory Control & Longitudinal Follow-Up Queue

[7]

[8]

- Doctor notices automated stock alert: *"⚠ IFA Tablets at 42% threshold – Reorder required"* [\[7\]](#).
- Follow-up system schedules automated post-discharge checklist for ASHA Shantha Devi [\[8\]](#).

PHASE 4: District Health Officer (THO) Governance & Analytics

Step 12, 13 & 14: Heatmap Analytics, Supply Chain & Audit

[7] [8] [9]

- **District Heatmap:** Taluk Health Officer reviews Tumakuru map (Pavagada: 29 high-risk cases, Koratagere: maternal spike).
- **Supply Chain Intervention:** Dispatches centralized medicine shipments to PHCs facing stockouts.
- **Clinical Audit:** Measures referral conversion efficiency (% of red flags reaching care facility).

🎬 3-Minute Live Hackathon Demo Script

"Judges, watch how ArogyaBandhu saves a rural mother's life in 180 seconds flat."

TIME	DEMONSTRATION ACTION ON SCREEN	FEATURES VERIFIED
0:00	Open ASHA Mobile App in airplane mode (offline indicator glowing, Kannada UI).	[11] [12]
0:20	Enter patient "Lakshmi Bai", 34w pregnancy, BP 162/104, severe swelling.	[3] [10] [14]
0:50	Click "Assess Risk" → Instant ● RED HIGH RISK alert appears on device.	[3]
1:00	Generate Digital QR Referral Slip with clinical summary.	[5]
1:15	Disable airplane mode → Automatic visual sync banner triggers: " ✓ Synced!"	[11]
1:30	Switch to PHC Doctor Portal → Lakshmi Bai referral lands at top of priority queue.	[2] [6]
1:50	Doctor clicks "108 Ambulance SOS" → Dispatch simulation confirmed.	[13]
2:15	Show Drug Stock Tracker with low IFA inventory warning.	[7]
2:30	Switch to THO District Dashboard showing Tumakuru taluk risk heatmap.	[9]
2:45	Click Language Toggle → Instant flawless flip to full ಕನ್ನಡ UI.	[12]
3:00	Wrap up: All 14 official features fully functional and validated!	ALL [1]–[14]

Anticipated Judges' Questions & High-Impact Answers

Q1: What makes this different from existing portals like RCH or Poshan Tracker?

Existing government portals are **retrospective compliance reporting tools** (data entered days later for administrative records). ArogyaBandhu is a **point-of-care clinical decision support system** that guides the health worker during the physical encounter, triages risk on the spot, and operates 100% offline.

Q2: How does the app guarantee zero data loss during prolonged offline periods?

We utilize browser **IndexedDB transactional persistence** with a queued synchronization state machine. Encrypted payload chunks remain safely stored on the device across browser restarts and power cycles until an HTTP 200 OK acknowledgment is received from the PHC gateway.

Q3: How is patient privacy maintained in accordance with national health standards?

We implement **ABDM (Ayushman Bharat Digital Mission) compliance** using 14-digit ABHA tokens. Data at rest is encrypted, and strict Role-Based Access Control (RBAC) ensures ASHAs only access assigned village households, doctors view facility referrals, and district officers receive de-identified epidemiological aggregations.

Q4: How does this strengthen rather than replace the existing public health apparatus?

ArogyaBandhu acts as a digital nervous system strictly respecting the National Health Mission (NHM) tiered hierarchy: **Sub-Centre ASHA → PHC Medical Officer → Taluk/District Hospital → 108 Arogya Kavacha**. It removes friction without requiring any bureaucratic restructuring.

Technology Architecture & Implementation Stack

LAYER	CHOSEN TECHNOLOGY	STRATEGIC RATIONALE
Frontend UI	HTML5 + Modern CSS3 + Vanilla JavaScript	Zero bundler overhead, instant cold start, 100% runtime reliability on low-spec phones.
Offline Database	Browser IndexedDB + LocalStorage Fallback	Structured schema storage with persistent local indexing and ACID transactions.
Referral QR Engine	Client-side QRCode.js	Instant vector QR generation on-device without cloud network roundtrips.
Analytics & Heatmaps	Chart.js + Responsive SVG Geometries	Lightweight, GPU-accelerated visualization for taluk epidemiology.
Localization	JSON Dictionary Engine (English / Kannada)	Zero latency language switching with native Noto Sans Kannada rendering.

